Team Meeting Minutes **Date: Sunday 26 April 2026**

Agenda item: Project kick-off - review Assignment 2B requirements and divide tasks

Discussion:

The team reviewed the Assignment 2B brief, rubric, and supporting resources including the SCATS dataset and Traffic Flow to Travel Time Conversion document. Task requirements were discussed: data processing, three ML models (LSTM, GRU, CNN-LSTM), travel time estimation, integration with Part A search algorithms, GUI, and report.

Conclusions:

All three members agreed on scope and timeline. Task division established: Saniru to lead data processing and integration; Haresh to lead LSTM and GRU implementation; Manula to lead CNN-LSTM and model evaluation. All members to contribute to report.

Action items	**Person**	**Deadline**
✓ Set up GitHub repository and project folder structure	All members	28 April 2026
✓ Read Traffic Flow to Travel Time Conversion document	All members	28 April 2026
✓ Begin data_processor.py implementation	Saniru	1 May 2026

Team Meeting Minutes **Date: Sunday 4 May 2026**

Agenda item: Data processing progress review and ML model architecture planning

Discussion:

Saniru demonstrated the data_processor.py pipeline - successful extraction of 4,192 rows from the SCATS XLS file, cleaning, MinMaxScaler normalisation, and sliding-window sequence generation (319,071 training sequences, sequence length 12). Haresh and Manula confirmed LSTM and GRU architecture designs: 64/32 stacked units, dropout 0.2, Adam optimiser.

Conclusions:

Data pipeline confirmed working correctly. 80/20 chronological train/test split agreed. Scaler to be fitted on training data only to prevent data leakage. Model architectures finalised.

Action items	**Person**	**Deadline**
✓ Implement LSTM model in train_models.py	Haresh	8 May 2026
✓ Implement GRU model in train_models.py	Haresh	8 May 2026
✓ Implement CNN-LSTM model	Manula	8 May 2026
✓ Begin travel_time.py implementation	Saniru	8 May 2026

Team Meeting Minutes **Date: Sunday 11 May 2026**

Agenda item: ML model implementation review and travel time conversion verification

Discussion:

Haresh presented LSTM and GRU implementations with stacked architecture, dropout, and early stopping callbacks. Manula presented CNN-LSTM design: two Conv1D layers followed by MaxPooling and LSTM. Saniru demonstrated travel_time.py implementing the quadratic flow-speed formula (A=-1.4648375, B=93.75) with 60km/h speed limit and 30-second intersection delay. Self-test verified 150.0 seconds at zero flow for a 2km segment.

Conclusions:

All three ML models implemented correctly. Travel time formula verified against the VicRoads specification. Team agreed to run initial training locally to verify convergence before next meeting.

Action items	**Person**	**Deadline**
✓ Run full model training and record MAE/RMSE/MAPE results	All members	14 May 2026
✓ Begin route_finder.py with graph construction	Saniru	14 May 2026
✓ Draft Introduction and Features sections of report	Manula	14 May 2026

Team Meeting Minutes **Date: Sunday 18 May 2026**

Agenda item: Route finder implementation and initial training results review

Discussion:

Saniru demonstrated route_finder.py: Boroondara SCATS graph built with 40 nodes and 100 directed edges. Connected-component bridging added to handle two disconnected site clusters. Yen's k-shortest paths with A* tested on O=2000 to D=3002, returning 5 routes of 19.5 to 20.4 minutes. Training results shared: CNN-LSTM best (MAE 13.90, RMSE 21.12, MAPE 28.16%), LSTM second, GRU third.

Conclusions:

Route finding confirmed working for the assignment reference example. CNN-LSTM confirmed as best performing model. Team agreed to proceed to GUI development.

Action items	Person	Deadline
✓ Implement full tkinter GUI (gui.py)	Saniru	22 May 2026
✓ Draft Insights and Testing sections of report	Haresh	22 May 2026
✓ Complete model evaluation write-up and result plots	Manula	22 May 2026

Team Meeting Minutes **Date: Sunday 25 May 2026**

Agenda item: GUI completion, testing verification, and full report review

Discussion:

Saniru demonstrated the completed GUI: origin/destination dropdowns, time-of-day picker, model selector, k-routes spinner, routes panel, Traffic Chart tab, and Model Metrics tab. All 12 test cases verified and documented. Full report draft reviewed by all members - Introduction, Features/Bugs/Missing, Testing, Insights, Research, Conclusion, and References sections all checked. Minor edits made to Section 5 analysis and Section 7 conclusion.

Conclusions:

GUI fully functional and tested. All 12 test cases pass. Report reviewed and approved by all members. README.md confirmed complete. All files committed to GitHub.

Action items	**Person**	**Deadline**
✓ Add screenshots to report placeholders	Saniru	29 May 2026
✓ Final submission to Canvas - all members individually	All members	30 May 2026
✓ Complete and sign contribution table	All members	30 May 2026

Team Meeting Minutes **Date: Friday 29 May 2026**

Agenda item: Final submission preparation and sign-off

Discussion:

Final pre-submission check completed. Full code package verified (data_processor.py, train_models.py, travel_time.py, route_finder.py, gui.py, search.py, config.json, README.md). Report PDF reviewed for page count, figure captions, header/footer, and APA references. Contribution table completed and signed. GitHub repository confirmed read-only access for OLA.

Conclusions:

All deliverables confirmed ready for submission. Each team member to individually submit to Canvas by 11:59 pm AEST Saturday 30 May 2026.

Action items	**Person**	**Deadline**
✓ Submit report PDF and code zip to Canvas	All members	30 May 2026
✓ Verify GitHub repo is accessible to OLA	Saniru	30 May 2026